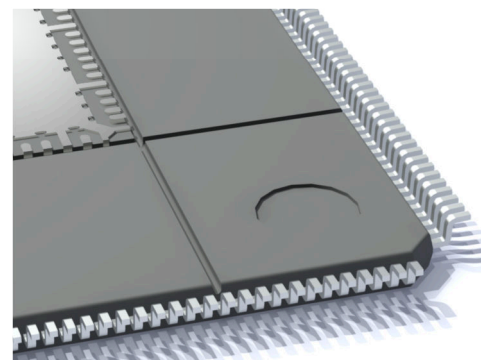
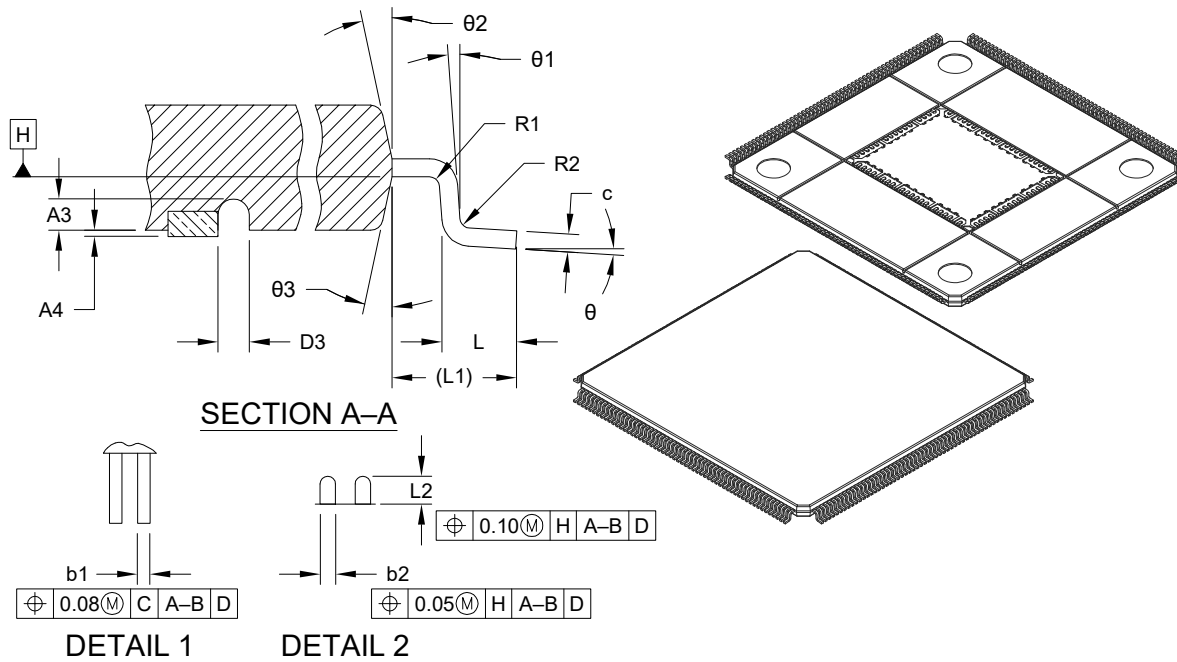


Note: For the most current package drawings, please see the Microchip Packaging Specification located at <http://www.microchip.com/packaging>



302-Lead Fusion Thin Quad Flat Pack (3HC) - 24x24x1.0 mm Body [FSTQFP] With Inner Terminals and 11.40x8.40 mm Exposed Pad; Microsemi Legacy Package

Note: For the most current package drawings, please see the Microchip Packaging Specification located at <http://www.microchip.com/packaging>



Units		MILLIMETERS		
Dimension Limits		MIN	NOM	MAX
Number of Terminals	N	302		
Number of Outer Terminals	—	224		
Number of Inner Terminals	—	78		
Outer Terminal Pitch	e1	0.40 BSC		
Inner Terminal Pitch	e2	0.50 BSC		
Overall Height	A	-	-	1.20
Molded Package Standoff	A1	0.00	0.051	0.10
Molded Package Thickness	A2	0.95	1.00	1.05
Groove Depth	A3	-	-	0.25
Inner Terminal Standoff	A4	-0.05	0.00	0.05
Overall Length	D	26.00 BSC		
Molded Package Length	D1	24.00 BSC		
Exposed Pad Length	D2	8.30	8.40	8.50
Groove Width	D3	-	-	0.40
Edge to Groove Center	D4	6.94 REF		
Overall Width	E	26.00 BSC		
Molded Package Width	E1	24.00 BSC		

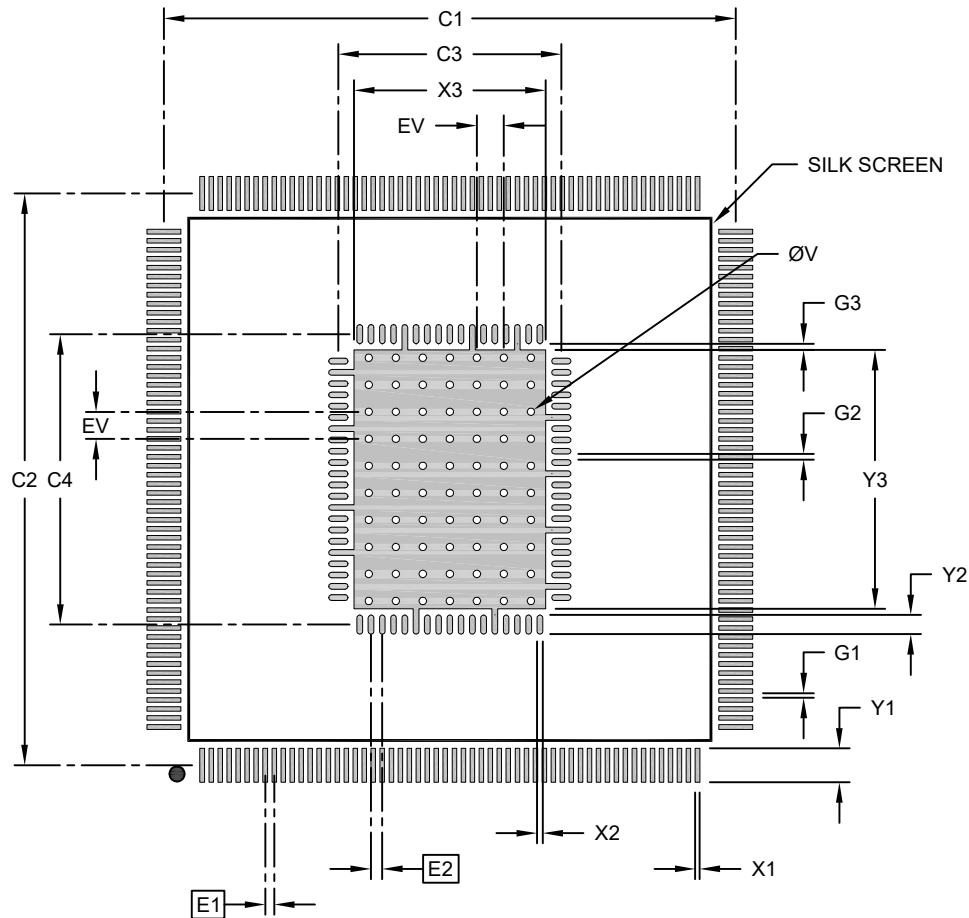
Units		MILLIMETERS		
Dimension Limits		MIN	NOM	MAX
Exposed Pad Width	E2	11.30	11.40	11.50
Edge to Groove Center	E4	5.44 REF		
Inner Term. Center to Center	D5	9.50 BSC		
Inner Term. Center to Center	E5	12.50 BSC		
Outer Terminal Width	b1	0.13	0.18	0.23
Inner Terminal Width	b2	0.17	0.22	0.27
Outer Terminal Thickness	c	0.14	-	0.20
Outer Terminal Length	L	0.45	0.60	0.75
Outer Terminal Footprint	L1	1.00 REF		
Inner Terminal Length	L2	0.30	0.40	0.50
Lead Bend Radius	R1	0.08	-	-
Lead Bend Radius	R2	0.08	-	0.20
Foot Angle	θ	0°	3.5°	7°
Lead Angle	θ1	0°	-	-
Mold Draft Angle	θ2	11°	12°	13°
Mold Draft Angle	θ3	11°	12°	13°

Notes:

- Pin 1 visual index feature may vary, but must be located within the hatched area.
- Dimensioning and tolerancing per ASME Y14.5M
BSC: Basic Dimension. Theoretically exact value shown without tolerances.
REF: Reference Dimension, usually without tolerance, for information purposes only.

302-Lead Fusion Thin Quad Flat Pack (3HC) - 24x24x1.0 mm Body [FSTQFP] **With Inner Terminals and 11.40x8.40 mm Exposed Pad; Microsemi Legacy Package**

Note: For the most current package drawings, please see the Microchip Packaging Specification located at <http://www.microchip.com/packaging>



RECOMMENDED LAND PATTERN

Units		MILLIMETERS		
Dimension Limits		MIN	NOM	MAX
Outer Terminal Pitch	E1		0.40 BSC	
Inner Terminal Pitch	E2		0.50 BSC	
Outer Terminal Pad Spacing	C1		25.40	
Outer Terminal Pad Spacing	C2		25.40	
Inner Terminal Pad Spacing	C3		9.90	
Inner Terminal Pad Spacing	C4		12.90	
Outer Terminal Pad Width (Xnn)	X1			0.20
Outer Terminal Pad Length (Xnn)	Y1			1.50

Units		MILLIMETERS		
Dimension Limits		MIN	NOM	MAX
Inner Terminal Pad Width (Xnn)	X2			0.25
Inner Terminal Pad Length (Xnn)	Y2			0.85
Center Pad Width	X3			8.50
Center Pad Length	Y3			11.50
Outer Term. Pad to Term. Pad (Xnn)	G1	0.20		
Inner Term. Pad to Term. Pad (Xnn)	G2	0.25		
Term. Pad to Center Pad (Xnn)	G3	0.28		
Thermal Via Diameter	V		0.33	
Thermal Via Pitch	EV		1.20	

Notes:

- Dimensioning and tolerancing per ASME Y14.5M
BSC: Basic Dimension. Theoretically exact value shown without tolerances.
- For best soldering results, thermal vias, if used, should be filled or tented to avoid solder loss during reflow process

Microchip Technology Drawing C04-27027 Rev A